

House Price Prediction using Machine Learning

This project focuses on building a machine learning regression model to predict house prices using housing-related features. The project includes data preprocessing, feature engineering, model training, evaluation, and prediction.

Exploratory Data Analysis (EDA) was performed to understand the dataset structure, feature relationships, and missing values. Visualizations such as scatter plots were used to analyze relationships between housing features and price.

Categorical variables were encoded using Label Encoding to make the data compatible with machine learning algorithms. The dataset was then split into training and testing sets for model evaluation.

Three regression algorithms were trained and compared:

1. Linear Regression
2. Random Forest Regressor
3. Gradient Boosting Regressor

The models were evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Residual analysis was also performed to analyze prediction errors and model performance.

Gradient Boosting Regressor achieved the best performance and was selected as the final model. The trained model was saved using Joblib and tested with sample prediction input.

This project demonstrates a complete beginner-level machine learning regression workflow including preprocessing, visualization, model training, evaluation, residual analysis, and model persistence.